

MSFASA-3DNet: Multi-Scale Feature Aggregation and Spatial Attention for 3-D Object Detection

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Abstract—Perception is the most important component in autonomous vehicles (AVs) and robots. LiDAR-based 3-D object detection (3D-OD) provides accurate spatial information for precise object localization. Traditional image-based 3D-OD methods struggle with depth estimation, making them less reliable for 3-D scene understanding. The existing LiDAR-based model uses single or two-stage convolutional networks to perform 3D-OD tasks. The existing models are computationally expensive with complex backbones. We propose an enhanced backbone with multiscale feature extraction and spatial attention to improve LiDAR-based 3D-OD. Our backbone is computationally effective as we have reduced almost 40% of the parameters of existing backbones and achieved competitive accuracy compared to existing models. Multiscale feature extraction captures object details at different resolutions to provide fine and coarse features. Spatial attention concentrates on important regions and suppresses background noise to generate more accurate spatial detections. Our backbone integrates multiscale feature aggregation and spatial attention mechanisms to improve detection accuracy. Our model uses the standard single-stage and grid-based 3D-OD pipeline with our proposed backbone. We conducted extensive ablation experiments to validate our proposed backbone's efficiency. Our model outperforms existing models by being lightweight, which makes it highly suitable for real-time applications. We evaluated our model using the KITTI benchmark. Our results are featured on the official leaderboard of KITTI.

Impact Statement—This proposed model presents a practical step forward in 3D-OD using LiDAR data for AVs and robotics. By designing a lighter and more efficient model backbone, the work significantly reduces computational complexity without sacrificing accuracy. The use of multiscale feature extraction helps capture both fine and broad details. The spatial attention focuses the model on the most important areas of a scene. This makes the system faster, more accurate, and well-suited for real-time applications where speed and reliability are crucial.

The model has been validated through strong performance on benchmark datasets.

Index Terms—3-D object detection (3D-OD), autonomous driving, convolutional neural networks (CNNs), deep learning, KITTI benchmark, LiDAR, multiscale feature extraction, perception, point cloud processing, spatial attention, voxel-based methods.

I. INTRODUCTION

PERCEPTION systems play a key role in autonomous vehicles (AVs) and robotics. Perception helps in understanding the environment by detecting objects, estimating distances, and interpreting the environment [1], [2], [3]. Object detection (OD) is one of the most important tasks in perception. The traditional 2-D object detection (2D-OD) provides only 2-D box information, which is not sufficient for AVs. The 3-D object detection (3D-OD) provides the 3-D box attributes location (x, y, z) , dimension (l, h, w) , and direction θ , which makes 3D-OD most suitable for AVs. The image-based 3D-OD is limited by inaccurate depth prediction [4], [5]. Unlike images, LiDAR is most prominent in terms of depth capturing, which makes LiDAR modality a primary choice for the 3D-OD for AVs. A LiDAR-based 3D-OD model takes raw point cloud data as input and generates the 3-D bounding boxes around the detected objects. Fig. 1 provides a visual preview of the proposed 3D-OD model.

LiDAR has emerged as a strong modality for 3D-OD with its precise long-range depth capturing. The point cloud data generated by the LiDAR is used for 3D-OD. Unlike cameras, LiDAR is not affected by lighting and weather conditions. This ensures reliable performance in both day and night scenarios. The high-resolution point cloud data provide detailed spatial understanding in 3-D space, which is essential for autonomous navigation and decision-making. LiDAR-based 3D-OD has gained a lot of attention from vision researchers in recent years.

The existing methods process LiDAR point clouds in point-based or grid-based techniques. Point-based methods [6], [7], [8], [9], [10], [11] operate directly on the raw point cloud to avail rich spatial information. The sparse and unstructured properties of point cloud data have higher computational requirements, making point-based methods a little expensive. There is a challenge to balance the complexity and accuracy. The grid-based approaches [12], [13], [14], [15], [16], [17], [18] convert raw point clouds into structured grids. The grid-based approach allows the processing of only voxels with data points and skips

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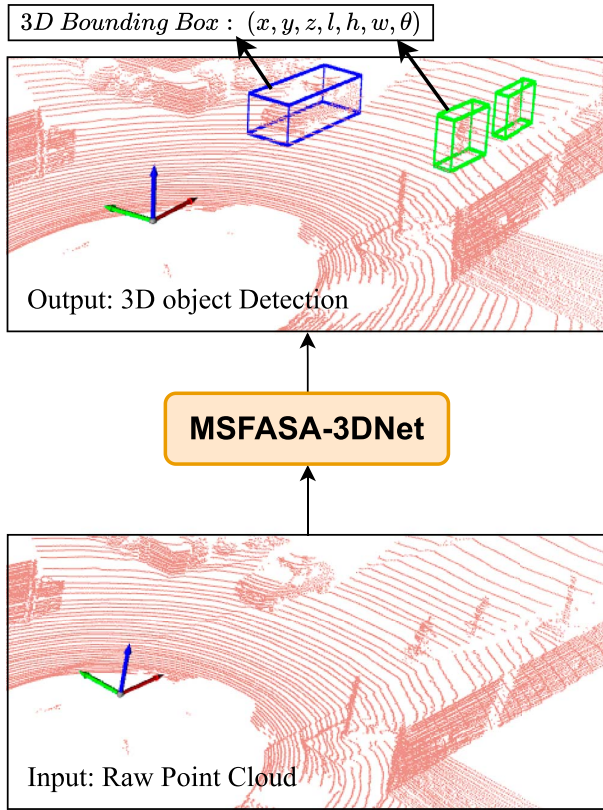


Fig. 1. Visual preview of our 3D-OD with our proposed model. The input to the model is raw point cloud data. The output scene is rendered with 3-D bounding boxes predicted by the proposed model.

the processing of empty voxels. The grid-based encoding is efficient and easy to process using convolutional neural networks (CNNs). Fusion approaches [19], [20], [21], [22] fuse the strengths of both voxel-based and point-based techniques to achieve a balance between efficiency and accuracy. The fusion approach might require additional operations, which makes the model heavier and more complex. Our proposed approach uses the grid-based approach [13], [23] for point cloud processing.

The 3D-OD models can be implemented in single or two stages. The two-stage models have an additional stage for ROI pooling or fusion practices. The two-stage models offer better accuracy but come with a computational cost. The balance between complexity and detection accuracy is the most important challenge in LiDAR-based 3D-OD. The single-stage models are easy to implement and deploy in real-time applications. Fig. 3 shows a standard single-stage and grid-based 3D-OD pipeline. We primarily focused on reducing the complexity of the backbone in the model.

Our work focuses on enhancing LiDAR-based 3D-OD. We propose a backbone that integrates spatial attention and multiscale feature extraction. Spatial attention helps in capturing important regions in the scene by focusing on significant areas while suppressing irrelevant background noise. Multiscale features improve detection across objects of different sizes, ensuring robust performance in varied environments. Unlike conventional hierarchical backbones [13], [19] or encoder-decoder structures, our approach captures multiscale context

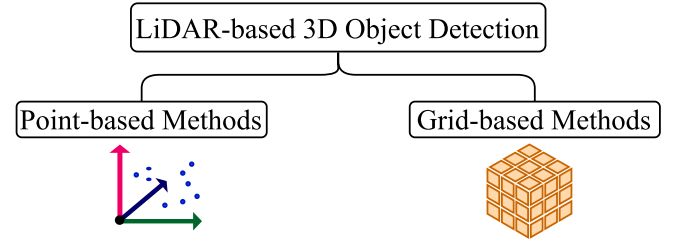


Fig. 2. Taxonomy of LiDAR-based 3D-OD models. The models are categorized into point-based and grid-based approaches.

through parallel receptive fields without introducing additional resolution levels. Spatial attention is applied with minimal computational overhead. The proposed modules are tightly integrated in a sequential flow that allows refinement of aggregated features. Our approach aims to improve accuracy while keeping computations efficient. We reduced 40% parameters compared to the existing backbones [13], [19], [23]. Our model achieved competitive accuracy while being lightweight. This makes it well suited for real-time applications in autonomous systems.

The major contributions of our proposed model are as follows.

- 1) In this article, we propose a novel backbone that incorporates multiscale context extraction, dual-branch feature refinement (DBFR), and spatial attention in a unified design.
- 2) We propose an end-to-end 3D-OD model, MSFASA-3DNet: multi-scale feature aggregation and spatial attention in LiDAR-based 3D-OD network.
- 3) We perform detailed ablation studies to validate the effect of each proposed module and evaluate their contributions toward overall performance.
- 4) We evaluated our model on the KITTI [24] benchmark dataset for the car and cyclist classes. Our model is featured in the leaderboard table on the official KITTI [25] website.

The article is structured in the following order. Section I provides the introduction, objective, challenges, and our proposal. Section II illustrates a detailed review of the existing methods. Section III presents our proposed method with a clear visual representation. Section IV provides the details about the experiments, model results, comparisons, and ablation studies. Section V presents the conclusion with remarks.

II. RELATED WORK

LiDAR-based 3D-OD gained a lot of attention among the computer vision community. There are wide categories of approaches to address the challenges in 3D-OD. The primary approaches are point based and grid based, based on the way models process the raw point cloud data. Fig. 2 shows the same in a graphical representation. The existing works are further classified into single and two-stage networks based on the number of stages in the networks.

A. Point-Based Methods

The point-based methods directly process raw point clouds without requiring transformation into voxel grids. PointNet [10] introduced a deep learning framework that directly operates on unstructured point cloud data without any preprocessing. The PointNet utilizes multi-layer perceptrons (MLPs) and symmetric aggregation functions. PointNet++ [11] extended PointNet by integrating hierarchical feature learning through set abstraction layers. Both models enable rich local feature extraction. The point-based models [6], [26] use point-based approaches and try to balance the accuracy and computational efficiency. The PI-RCNN [8] uses raw point clouds to extract 3-D proposals and uses them for further fusion with other modalities. Point-based approaches are widely chosen by researchers who aim for full resolution and rich spatial information. The point-based models suffer from the unstructured and sparse nature of the LiDAR point cloud.

B. Grid-Based Methods

The grid-based methods convert point clouds into structured voxel grids. The VoxelNet [12] introduced this voxelization process and succeeded in the efficient encoding of the point cloud data. Following the advantages and success of this approach, many existing methods [13], [15], [16], [17], [18], [27], [28] have adopted this for point cloud processing. The PV-RCNN [19] and its variant PV-RCNN++ [29] use voxelization to fuse with point-based features to utilize rich features. Maff-Net [30] uses the VoxelNet itself as a backbone. The methods such as Voxel-RCNN [31] and SegVoxelNet [18] use a similar approach followed in VoxelNet [12]. The key benefit of utilizing voxel-based methods is to avoid processing empty voxels. These methods process only receptive fields with data points, which address the challenge of sparse point cloud data. Our proposed model also utilizes a grid-based approach to process the point cloud data.

C. Stages in Network

The 3D-OD models can have single or multiple stages. The single-stage networks [7], [12], [13], [14], [15], [16], [18], [27], [28], [30], [32] will have a 3-D feature extraction unit to extract the 3-D feature volumes and a 2-D backbone to generate proposals from the compressed 3-D features. The output feature map from the 2-D backbone is shared by the detection unit to generate the OD results. Fade3D [33] proposes a fast 3D-OD framework for LiDAR point clouds. Fade3D incorporates a lightweight input encoder and an efficient bird's eye view (BEV) backbone in a single stage. The two-stage models [6], [8], [9], [17], [19], [20], [29], [34], [35], [36], [37] are typically fusion based, which use multiple modalities. These methods use an additional stage dedicated to the proposal refinement or fusion. The existing two-stage methods [9], [19], [31] use region pooling, abstraction layers, and RGB data fusion to refine the proposal generated in the first stage. The additional refinement stages will raise the complexity but with improved accuracy. The challenge is to bring down the complexity and increase the

accuracy of the model. Our proposed model is a single-stage design and eliminates the expensive refinement layers. Fig. 3 illustrates the complete pipeline of our single-stage lidar-based model.

D. Multiscale Feature Aggregation

Processing LiDAR data effectively require capturing geometric features across varying object scales and sparsity levels. Traditional multiscale approaches such as feature pyramid networks (FPNs) [38] and atrous spatial pyramid pooling (ASPP) [39] have been explored for dense image tasks. However, directly applying them to sparse 3-D point clouds can cause inefficiencies due to mismatched spatial distributions. Instead of relying on hierarchical backbones or encoder-decoder structures [7], [12], [13], our method applies parallel receptive fields to capture multiscale spatial context more efficiently. We design a multiscale feature aggregation module specifically adapted for LiDAR, which enhances spatial understanding without significantly increasing complexity. These components are integrated into a single-stage, end-to-end architecture for LiDAR-based 3D-OD.

E. Attention in 3D-OD

Spatial attention helps LiDAR-based models to focus on key regions within a scene. Recent methods have explored the use of attention mechanisms [40], [41] to enhance OD by emphasizing informative regions. TANet [17] uses triple attention to extract more discriminating features. A recent method [42] has introduced feature pyramid structures and efficient attention mechanisms to improve perception performance in visual place recognition tasks. Recent advancements [43] in multimodal navigation have demonstrated the effectiveness of integrating LiDAR with attention mechanisms for robust scene understanding. Our approach integrates spatial attention tailored for LiDAR-based 3D-OD.

We are using the grid-based approach to process the point cloud data and generate the 3-D feature volumes [12], [13], [23]. The proposed model is a single-stage and completely end-to-end 3D-OD model.

III. PROPOSED MODEL

LiDAR-based 3D-OD relies on the backbone, which is a core learning unit in the pipeline. Our proposed backbone is integrated with multiscale aggregation and a spatial attention mechanism to boost detection accuracy. Our proposed model follows a standard pipeline as shown in Fig. 3. The first unit in the pipeline converts the raw point cloud into a voxel representation. The voxel data are processed to extract 3-D features, and then the features will be height compressed into a 2-D representation for further feature refinement. The proposed backbone will learn the spatial features, and these feature maps will be used by the detection head (DH) to generate 3D-OD results. The goal of our approach is to enhance spatial and contextual feature learning while maintaining computational efficiency.

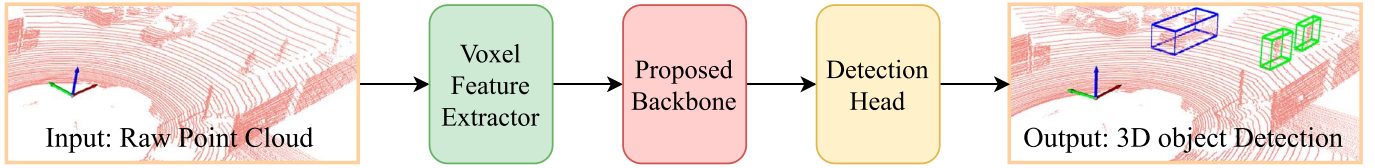


Fig. 3. Detection pipeline. The raw point cloud is converted into voxels, generating voxel feature volumes in the voxel feature extractor. These feature volumes are then compressed into a 2-D representation and fed into the proposed backbone for further processing. The final feature map from the 2-D backbone is utilized by the DH to predict 3-D bounding box attributes. The output scene is rendered with the detected objects.

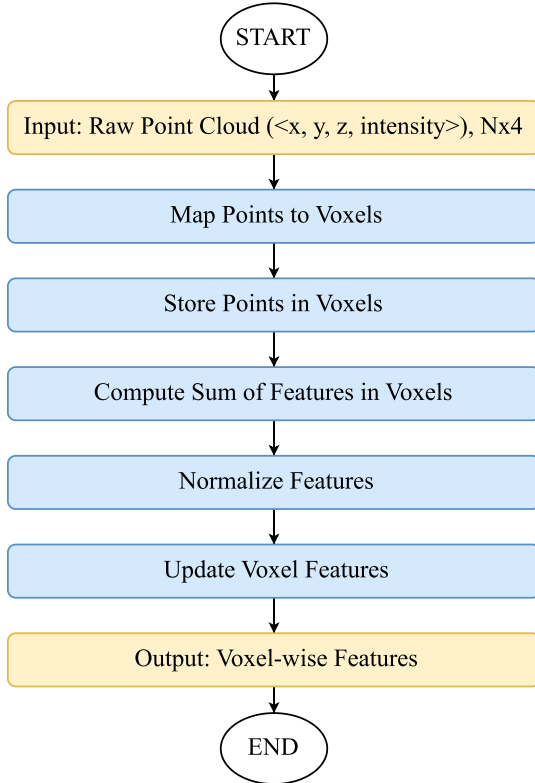


Fig. 4. Flowchart of the voxelization process: This diagram illustrates the step-by-step workflow for converting raw point cloud data into voxelwise features.

A. Voxel Feature Extractor

The role of this unit is to convert the raw point cloud into voxels and then process the voxels to generate the 3-D voxel feature volumes. The raw point cloud is transformed into voxels using widely adopted voxelization [12], [13], [19], [23], [31] approach. The voxelization process follows the implementation from OpenPCDet [23]. The detailed step-by-step process of voxelization is illustrated in Fig. 4. These voxels are then passed through a sparse 3-D convolutional backbone to extract voxel feature volumes [13], [23]. This is a very important step in capturing both geometric and structural information. The usage of sparse convolutions is introduced in the SECOND [13] to address the time complexity issue. We also used the same backbone to extract the 3-D feature volumes. The high-dimensional 3-D representation is compressed into a 2-D feature representation to serve as the input to our proposed 2-D backbone.

B. Proposed Backbone

Our 2-D backbone is designed to enhance feature representation through multiscale feature learning and spatial attention. These mechanisms are most suitable for extracting rich features from LiDAR-based representations. The proposed backbone consists of four key components: high-resolution feature encoder (HRFE), multiscale context aggregator (MSCA), DBFR, and context-aware spatial attention (CASA).

These components work together to extract meaningful features and capture multiscale contextual information. Fig. 5 shows the module-level overview of the backbone. Fig. 6 shows the component-level configuration details of the proposed backbone. The enhanced discriminative power leads to improved detection accuracy. We reduced $\sim 40\%$ of the parameters compared to the baseline models [13], [19]. Hence, the proposed novel backbone is lightweight and effective.

1) *HRFE*: The HRFE module is responsible for initial feature extraction. We preserved the spatial resolution to allow for the extraction of detailed contextual and spatial information. It consists of a sequential stack of convolutional layers with ReLU activation and batch normalization (BN). This block is designed for low-level feature encoding without compromising the structural details of the input representation.

The input feature map $F_0 \in \mathbb{R}^{C_{in} \times H \times W}$ is passed through a sequence of four convolutional blocks, each consisting of a 3×3 convolution with stride 1 and padding 1, followed by BN and ReLU activation.

The output of the HRFE module is defined as

$$F_{HRFE} = \psi_4 \circ \psi_3 \circ \psi_2 \circ \psi_1(F_0) \quad (1)$$

where $\psi_i(\cdot)$ denotes a single Conv-BN-ReLU operation at stage i and $\circ$ denotes function composition. Equation (1) represents the sequential flow of operations in the HRFE block. The resulting feature map $F_{HRFE} \in \mathbb{R}^{128 \times H \times W}$ is forwarded to the MSCA module.

2) *MSCA*: The MSCA is the key contribution of the backbone. The MSCA captures features at multiple receptive fields to learn rich contextual information. We employed three parallel convolutional branches with different kernel sizes: 1×1 , 3×3 , and 5×5 . This allows the network to extract both fine-grained and high-level contextual information. The outputs of these three branches are concatenated to form a rich multiscale representation. This process enhances the ability to recognize objects of varying sizes and orientations.

Given an input feature map $F_{HRFE} \in \mathbb{R}^{C \times H \times W}$ ($C = 128$), the MSCA module applies three parallel convolutional branches

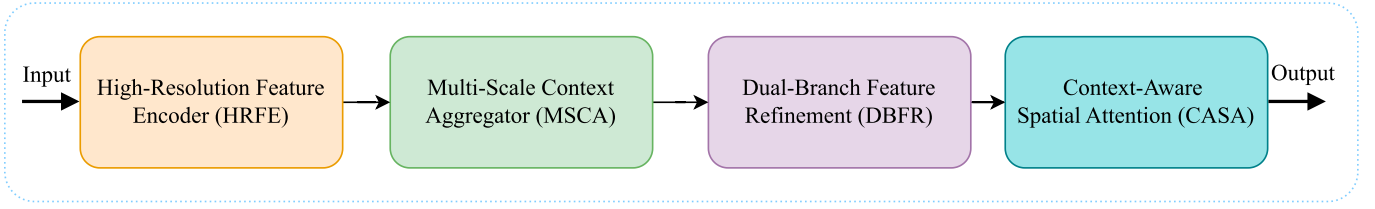


Fig. 5. Module-level architecture of the proposed backbone. The input features are processed through HRFE for initial feature encoding. MSCA captures features at multiple receptive fields to extract both fine-grained and high-level contextual information. DBFR refines multiscale representations through dual branch refinement. CASA enhances discriminative feature learning by applying a lightweight spatial attention mechanism. The final feature map is passed to the DH for 3D-OD.

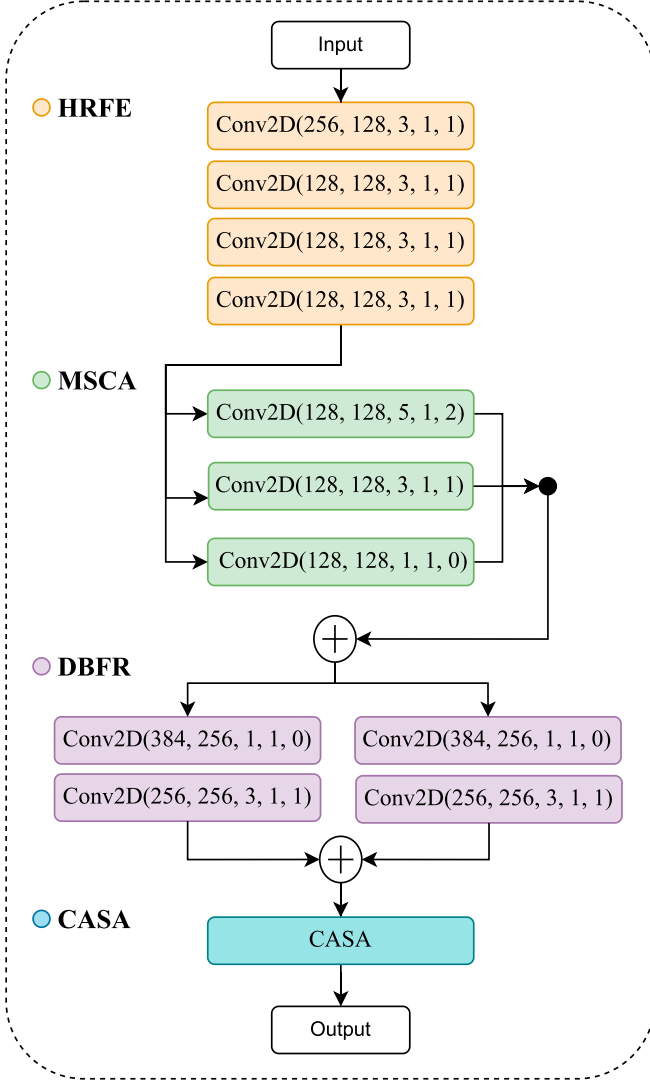


Fig. 6. Network architecture of the proposed backbone with configuration details. The parameters of the convolution operation are in the order of input channels, output channels, kernel size, stride, and padding.

with kernel sizes 1×1 , 3×3 , and 5×5 to capture contextual information at different scales. Each branch includes a convolution layer, followed by BN and a ReLU activation.

The outputs of the branches are computed as

$$F_{1 \times 1} = f_{1 \times 1}(F_{\text{HRFE}})$$

$$F_{3 \times 3} = f_{3 \times 3}(F_{\text{HRFE}})$$

$$F_{5 \times 5} = f_{5 \times 5}(F_{\text{HRFE}}) \quad (2)$$

where $f_{k \times k}(\cdot)$ represents a convolution with kernel size $k \times k$ followed by BatchNorm and ReLU.

These outputs are concatenated along the channel dimension, as defined in

$$F_{\text{MSCA}} = \text{Concat}(F_{1 \times 1}, F_{3 \times 3}, F_{5 \times 5}) \quad (3)$$

to form a rich multiscale feature representation. Equation (2) summarizes the operations of the three parallel branches, while (3) shows how they are fused to obtain the final aggregated feature map $F_{\text{MSCA}} \in \mathbb{R}^{3C \times H \times W}$.

3) *DBFR*: The DBFR module further refines the aggregated multiscale features generated by MSCA. Here, we apply two parallel transformation blocks for feature learning. These blocks operate separately on the multiscale feature representation and generate two feature maps. These generated feature maps are concatenated to maintain a well-channeled feature representation. This structure enables the model to capture complementary refinements of the same input representation.

Let $F_{\text{MSCA}} \in \mathbb{R}^{C' \times H \times W}$ be the input to the DBFR module, where $C' = 3C$ corresponds to the channel dimension after concatenation in MSCA. Each branch applies a 1×1 convolution followed by a 3×3 convolution, with intermediate BN and ReLU activations. The outputs from the two branches are computed as

$$F_1 = g_1(F_{\text{MSCA}}), \quad F_2 = g_2(F_{\text{MSCA}}) \quad (4)$$

where $g_1(\cdot)$ and $g_2(\cdot)$ each represent a sequential composition of 1×1 convolution, BN, ReLU activation, and 3×3 convolution. The outputs are concatenated to produce the final refined feature map

$$F_{\text{DBFR}} = \text{Concat}(F_1, F_2). \quad (5)$$

Equation (4) captures the dual-branch refinement process applied to the input features and outputs $F_{\text{DBFR}} \in \mathbb{R}^{C'' \times H \times W}$ ($C'' = 512$). The fusion in (5) enables the network to combine different perspectives of spatial and channel-level transformations before applying spatial attention in the subsequent stage.

4) *CASA*: This is one of the major contributions to the proposed backbone. The CASA is a lightweight spatial attention module that highlights the important regions that need to be detected. The spatial attention in our backbone is different from widely used convolutional block attention module (CBAM) [40]. We used learnable spatial convolution in the block and added a skip connection from the input. The hyperparameters

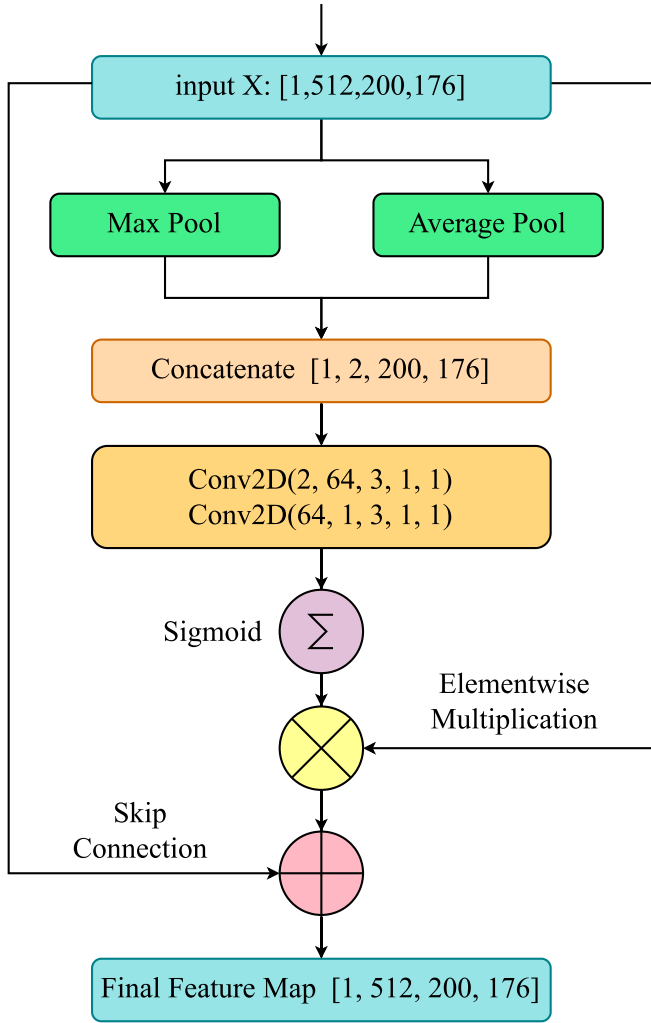


Fig. 7. CASA architecture: we applied max and average pooling along the channel dimension, followed by a convolutional block to generate the attention map. The attention is then applied to the input via elementwise multiplication and a skip addition.

are also different from those of the existing block. The complete design of CASA is illustrated in Fig. 7, including the configuration. The attention module first applies global max pooling and mean pooling along the channel dimension to capture essential contextual information. These pooled features are then processed through a learnable convolutional module to generate a spatial attention map. The final attention weights are applied to the feature map to enhance important regions while suppressing less relevant ones. The skip connection from the input helps mitigate gradient vanishing issues.

Given the input feature map $F_{\text{DBFR}} \in \mathbb{R}^{C'' \times H \times W}$, the attention mechanism first computes two spatial descriptors using channelwise global pooling operations

$$F_{\text{pool}} = \text{Concat}(\max_c(F_{\text{DBFR}}), \text{avg}_c(F_{\text{DBFR}})) \quad (6)$$

where $\max_c$ and $\text{avg}_c(\cdot)$ denote the maximum and average over the channel dimension, resulting in two maps of shape $\mathbb{R}^{1 \times H \times W}$. Their concatenation yields a two-channel attention descriptor $F_{\text{pool}} \in \mathbb{R}^{2 \times H \times W}$. This descriptor is passed through

a pair of convolutional layers and a sigmoid function to produce the attention map

$$M_{\text{att}} = \sigma(\text{Conv}_{3 \times 3}(\text{ReLU}(\text{Conv}_{3 \times 3}(F_{\text{pool}})))) \quad (7)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. The attention map $M_{\text{att}} \in \mathbb{R}^{1 \times H \times W}$ highlights spatially informative regions. Finally, the input features are reweighted and combined with a residual connection

$$F_{\text{CASA}} = F_{\text{DBFR}} \odot M_{\text{att}} + F_{\text{DBFR}} \quad (8)$$

where $\odot$ denotes elementwise multiplication broadcast across channels. Equation (6) computes the spatial descriptors, (7) generates the attention map, and (8) applies the attention-enhanced residual transformation. This operation helps the network focus on informative spatial regions while preserving the original feature structure.

The backbone processes input feature maps sequentially through HRFE, MSCA, DBFR, and CASA. The final feature map serves as an input to the DH. The DH is responsible for object classification and localization. The highlight of the proposed novel backbone is a $\sim 40\%$ reduction of parameters compared to baselines. The backbone is lightweight in structure but outperforms the existing single and two-stage models. The proposed backbone enhances the robustness and accuracy of 3D-OD with fewer parameters in LiDAR-based perception tasks. To support our backbone, we conducted various ablation experiments on the KITTI test set.

C. DH

The DH is responsible for producing the 3D-OD results. The tasks in the DH are classification and bounding box regression. The classification task is to predict the object category as car/cyclist. The regression task is to predict the bounding box attributes and direction. The prediction results include the precise location, size, and dimensions of the object in 3-D. The DH uses three convolutional layers for classification and regression of bounding box attributes and regression for direction [23]. The three convolutional layers share the same output feature map from the backbone.

D. Loss Function

The proposed model carries out two key tasks: classification for predicting the object class and regression for estimating the bounding box coordinates and direction. Therefore, the loss function used in our model is a combination of both losses.

1) *Classification Loss*: We adopt the focal loss introduced by Lin et al. in RetinaNet [44] for the classification task. This loss function and its variants have been successfully applied in various models [13], [19]. The classification loss is defined as shown in

$$L_{\text{cls}} = -(1 - p_t)^\gamma \log(p_t) \quad (9)$$

where p_t represents the model's predicted probability for the class with label t , γ is a focusing parameter that controls how quickly easy examples are down-weighted.

TABLE I
ABLATION STUDY SHOWING THE IMPACT OF INDIVIDUAL DATA
AUGMENTATION TECHNIQUES ON 3-D AND BEV DETECTION

Augmentation Type	3-D Detection		BEV Detection	
	Car (mAP)	Cyclist (mAP)	Car (mAP)	Cyclist (mAP)
No Augmentation	74.64	61.59	85.44	64.89
Flip Only	79.56	63.73	87.72	67.94
Rotation only	82.73	65.49	89.12	70.86
Scaling only	79.06	61.41	87.49	66.05
All three (ours)	83.41	70.37	90.21	74.52

Note: The highest value is in bold.

2) *Regression Loss*: We used the Smooth-L1 loss for the regression task. The regression loss is computed as shown in

$$L_{\text{reg}} = \sum_i \text{Smooth-L1}(\Delta x_i) \quad (10)$$

where Δx_i represents the difference between the estimated and the corresponding actual truth values. The predicted and target bounding box parameters are in the standard format $(x, y, z, h, l, w, \theta)$.

IV. EXPERIMENTS

A. Dataset Description

1) *KITTI Dataset*: The KITTI Vision Benchmark Suite [24], [25], [45] was developed by the Karlsruhe Institute of Technology and Toyota Technological Institute. KITTI is a widely used dataset for computer vision applications, particularly in autonomous driving research. KITTI provides various benchmarks such as 3-D, BEV, and 2-D. We conducted our experiments on the 3D-OD benchmark within this suite. The 3-D detection benchmark includes 7481 LiDAR samples in the training set and 7518 in the test set. Ground truth annotations are available only for the training samples. To evaluate model performance on the test set, submissions must be made to the KITTI evaluation server in accordance with its official policy. Existing works split the training dataset into separate training and validation subsets for experimental purposes [23].

B. Network Architecture

The proposed backbone follows the same parameter representation for convolution operation in the order of input channels, output channels, kernel size, stride, and padding. All convolution layers are followed by BN and a ReLU activation function, except for the last layer in the CASA block. The backbone processes features through a structured series of convolutional layers for feature extraction and refinement. Figs. 4, 6, and 7 show the full configuration with parameters. The network begins with the HRFE block feature extraction module, which consists of four consecutive 3×3 convolutional layers with 128 channels. A multiscale feature extraction module applies three parallel convolutional branches: 1×1 , 3×3 , and 5×5 convolution layers. The outputs of three scaled convolution branches are concatenated along the channel dimension and form a 384-channel representation. The concatenated features are then refined through two processing blocks in DBFR, each comprising a 1×1 convolution that reduces the channels from 384 to 256,

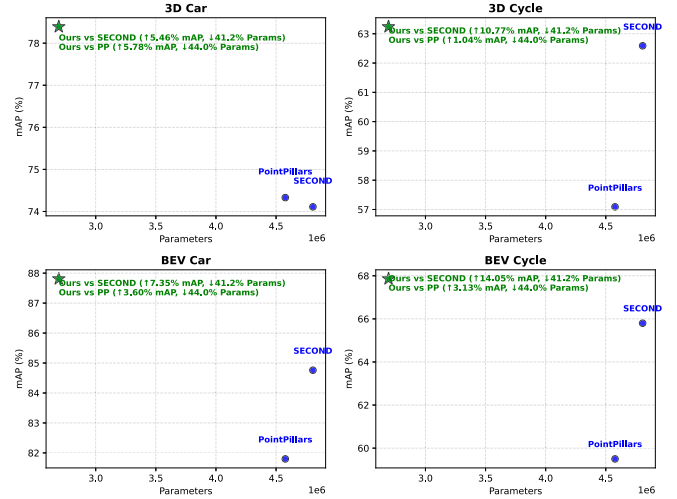


Fig. 8. Comparison of detection accuracy and model parameters for our proposed approach against strong baselines [13], [14]. Our method achieves higher detection accuracy with approximately 40% fewer parameters compared to SECOND and PointPillars, demonstrating its efficiency.

TABLE II
BENCHMARK RESULTS ACROSS DIFFERENT GPU PLATFORMS

Device	Inference Time (s)	FPS	Memory (MB)
Jetson Orin 8GB	0.0910	10.99	536.17
NVIDIA T400 2GB	0.1652	6.05	537.87
NVIDIA T400 4GB	0.1622	6.17	537.87
NVIDIA RTX A4000 16GB	0.0222	44.98	536.17
NVIDIA A100-PCIE-40GB	0.0059	168.46	536.17

followed by a 3×3 convolution maintaining 256 channels. The CASA applies max pooling and average pooling along the channel axis to generate a compressed feature map, which is then processed through two 3×3 convolutional layers with 64 and 1 channels, respectively. A sigmoid activation function produces an attention map, which is elementwise multiplied with the refined features to emphasize critical areas. Finally, the processed features are generated by maintaining a total of 512 channels for the next module.

C. Implementation Details

Our model is trained for 80 epochs with a batch size of 4 on the KITTI dataset. We used OpenPCDet [13], [19], [23], an open-source framework available on GitHub. We implemented experiments on an NVIDIA A4000 GPU running Ubuntu 20.04 LTS operating system. We used the Adam optimizer with the one-cycle learning rate (LR) policy. The LR is initialized at 0.01, and a weight decay of 0.01 is applied for regularization. The one-cycle policy is configured with a percentage start of 40% and an LR division factor of 10. LR decay occurs at epochs 35 and 45, reducing by a factor of 0.1. The minimum LR is clipped at 1×10^{-7} to prevent instability. The total training time for 80 epochs was 13.4 h. The training process was stable, and the model showed consistent convergence across epochs. The training loss decreased smoothly and stabilized around

TABLE III
QUANTITATIVE RESULTS ON THE *KITTI TEST SET* FOR THE *3D-OD BENCHMARK*

Model	Inference Time (s)	Car			Cyclist		
		Easy	Mod.	Hard	Easy	Mod.	Hard
VoxelNet (CVPR 2018) [12]	0.22	77.47	65.11	57.73	61.22	48.36	44.37
SECOND (MDPI 2018) [13]	0.04	83.13	73.66	66.2	70.51	53.85	46.90
AVOD-FPN (IROS 2018) [36]	0.08	81.94	71.88	66.38	64	52.18	46.61
PointPillars (CVPR 2019) [14]	0.02	79.05	74.99	68.30	75.78	59.07	52.92
SCNet (IEEE Access 2019) [15]	0.04	82.84	73.9	66.91	67.66	51.06	47.54
Fast Point RCNN (ICCV 2019) [34]	0.07	84.28	75.73	67.39	-	-	-
ARPNET* (Science China. Information Sciences 2019) [47]	0.08	84.69	74.04	68.64	74.21	58.2	52.13
Complexer-YOLO (CVPR Workshop 2019) [48]	0.06	55.63	49.44	44.13	28.36	23.48	22.85
SegVoxelNet (ICRA 2020) [18]	0.04	84.19	75.81	67.80	-	-	-
Voxel-FPN (MDPI 2020) [16]	0.02	85.64	76.70	69.44	-	-	-
Associate3ddet (CVPR 2020) [28]	0.06	85.99	77.40	70.53	-	-	-
TANET (AAAI 2020) [17]	0.04	83.81	75.38	67.66	73.84	59.86	53.46
PI-RCNN (AAAI 2020) [8]	0.10	84.37	74.82	70.03	-	-	-
WS3D* (ECCV 2020) [49]	0.10	80.99	70.59	64.23	-	-	-
RangeDet* (ICCV 2021) [32]	0.02	85.41	77.36	72.60	-	-	-
PFF3D (IEEE Access 2021) [26]	0.05	81.11	72.93	67.24	63.27	46.78	41.37
DASS* (WACV 2021) [50]	0.09	81.85	72.31	65.99	-	-	-
MAFF-Net (ICITS 2022) [30]	0.04	84.31	75.37	70.3	-	-	-
PointRGBNet (Automotive Engineering 2022) [51]	0.08	83.99	73.49	68.56	67.05	52.15	46.78
Harmonic PointPillar* (IEEE TVT 2023) [7]	0.01	82.26	73.96	69.21	-	-	-
EOTL (IEEE Sensors 2023) [6]	-	79.97	69.13	58.57	75.2	58.96	50.41
PCNet3D (CVIP 2024) [52]	0.05	83.22	73.58	68.19	74.15	57.28	51.02
Fade3D (IEEE TITS 2025) [33]	0.01	85.85	75.57	70.47	66.04	50.02	44.39
Ours	0.03	86.06	77.21	71.89	75.6	60.30	53.83

Note: Our method is officially listed on the KITTI leaderboard [25]. Bold numbers indicate the best performance in each column. Models marked with “*” have their results reported from the official KITTI website.

TABLE IV
QUANTITATIVE RESULTS ON THE *KITTI TEST SET* FOR THE *BEV DETECTION BENCHMARK*

Model	Inference Time (s)	Car			Cyclist		
		Easy	Mod.	Hard	Easy	Mod.	Hard
VoxelNet (CVPR 2018) [12]	0.22	89.35	79.26	77.39	66.7	54.76	50.55
SECOND (MDPI 2018) [13]	0.04	88.07	79.37	77.95	73.67	56.04	48.78
AVOD-FPN (IROS 2018) [36]	0.08	88.53	83.79	77.9	68.06	57.48	50.77
PointPillars (CVPR 2019) [14]	0.02	88.35	86.1	79.83	79.14	62.25	56
SCNet (IEEE Access 2019) [15]	0.04	88.87	86.46	78.81	72.25	56.93	50.88
Fast Point RCNN (ICCV 2019) [34]	0.07	88.03	86.1	78.17	-	-	-
ARPNET* (Science China. Information Sciences 2019) [47]	0.08	90.06	86.81	79.41	82.32	66.39	58.80
Complexer-YOLO (CVPR Workshop 2019) [48]	0.06	74.23	66.07	65.7	36.12	30.16	26.01
SegVoxelNet (ICRA 2020) [18]	0.04	88.62	86.16	78.68	-	-	-
Voxel-FPN (MDPI 2020) [16]	0.02	92.75	87.21	79.82	-	-	-
Associate3ddet (CVPR 2020) [28]	0.06	91.4	88.09	82.96	-	-	-
TANET (AAAI 2020) [17]	0.04	91.58	86.54	81.19	79.16	63.77	56.21
PI-RCNN (AAAI 2020) [8]	0.10	91.44	85.81	81	-	-	-
WS3D* (ECCV 2020) [49]	0.10	90.96	84.93	77.96	-	-	-
RangeDet* (ICCV 2021) [32]	0.02	90.93	87.67	82.92	-	-	-
PFF3D (IEEE Access 2021) [26]	0.05	89.61	85.08	80.42	72.67	55.71	49.58
DASS* (WACV 2021) [50]	0.09	91.74	85.85	80.97	-	-	-
MAFF-Net (ICITS 2022) [30]	0.04	90.47	87.47	82.69	-	-	-
PointRGBNet (Automotive Engineering 2022) [51]	0.08	91.39	85.73	80.68	73.09	57.59	51.78
Harmonic PointPillar* (IEEE TVT 2023) [7]	0.01	90.89	87.28	82.54	-	-	-
EOTL (IEEE Sensors 2023)* [6]	-	89.1	83.14	71.41	81.44	65.76	56.47
PCNet3D (CVIP 2024) [52]	0.05	90.09	86.54	81.43	78.52	62.48	55.54
Fade3D (IEEE TITS 2025) [33]	0.01	93.53	87.85	82.75	71.51	55.00	48.65
Ours	0.03	91.35	87.7	84.38	79.61	65.12	58.84

Note: Our method is officially listed on the KITTI leaderboard [25]. Bold numbers indicate the best performance in each column. Models marked with “*” have their results reported from the official KITTI website.

0.240 by the final epoch. We followed the hyperparameter configuration from the official OpenPCDet [23] repository, which has been extensively validated for LiDAR-based 3D-OD tasks

on the KITTI dataset. These hyperparameters have been shown to produce stable convergence and competitive performance across many existing works.

We apply various data augmentation strategies [19] to enhance the performance of our model. We used augmentation techniques, including random world flipping along the x-axis, random world rotation within $[-0.785, 0.785]$ radians, and random world scaling within a range of $[0.95, 1.05]$. These augmentations ensure a diverse and robust training dataset. A quantitative analysis of these augmentations and their impact on performance is presented in Table I. The reported results highlight the impact of our augmentation selection.

D. Evaluation Metrics

The official evaluation metric of KITTI [24] is average precision (AP) computed with 40 recall positions. We evaluated our model in the car and cyclist classes. The IoU threshold for cars is 70%, and for cyclists, it is 50%. Considering the height of the 3-D bounding box, occlusion and truncation values, each class is divided into easy, moderate, and hard difficulty levels. The formula for AP [46] is as follows:

$$AP = \frac{1}{N} \sum_{i=1}^N P(R_i) \quad (11)$$

where R_i represents the recall value at a specific i , and $P(R_i)$ is the precision at that particular recall value. N is the number of recall positions, 40 as per KITTI.

The mAP is the mean of APs across all classes in the dataset [46]

$$mAP = \frac{1}{C} \sum_{c=1}^C AP_c \quad (12)$$

where C is the total number of classes, and AP_c is the AP value for class c .

E. Performance Versus Model Complexity

The main challenge in LiDAR-based 3D-OD is to balance model parameters and accuracy. We presented the effect of parameter reduction on car and cyclist classes for 3-D and BEV detection tasks. The selected baselines are strong, and many others use similar/same backbones in their framework. Our backbone has 2.69M parameters, showing a $\downarrow 41.2\%$ reduction compared to SECOND (4.58M) and $\downarrow 44.0\%$ compared to PointPillars (4.81M). We achieved higher detection accuracy with $\sim 40\%$ fewer parameters compared to SECOND and PointPillars. Fig. 8 shows the parameter vs detection accuracy for our proposed model compared to strong baselines [13], [14]. Our model has achieved competitive accuracy in both car and cyclist classes with a lightweight design. We evaluated the run-time performance of the proposed backbone across a range of hardware platforms to assess its real-time inference capability. Table II reports benchmark comparisons, including inference time, frames per second (FPS), and memory usage, on various GPUs. These results demonstrate that the proposed model maintains low-inference time and moderate-memory consumption across diverse deployment scenarios.

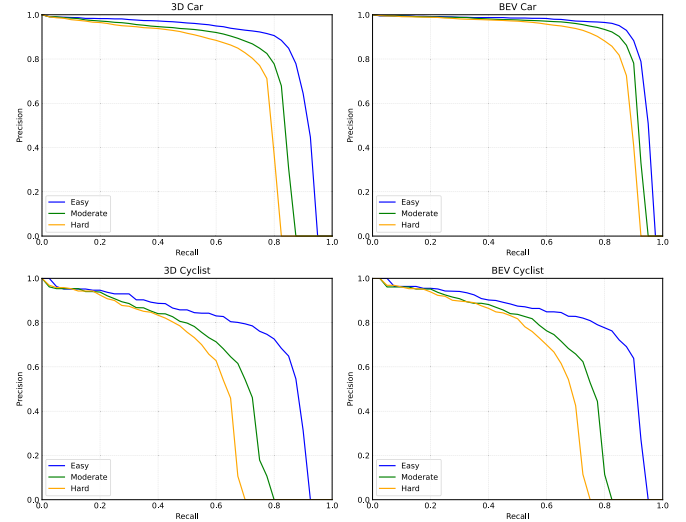


Fig. 9. Precision–recall (PR) curves: Visualization of the tradeoff between precision and recall across different confidence thresholds. The curves show our model’s results for each class in 3-D and BEV.

TABLE V
ABLATION STUDY ON HRFE FOR CAR AND CYCLIST DETECTION
UNDER 3-D AND BEV REPRESENTATIONS ON THE KITTI DATASET

Model	Car (AP)			Cyclist (AP)		
	Easy	Mod.	Hard	Easy	Mod.	Hard
Without HRFE (3-D)	85.61	74.53	71.18	77.02	56.73	52.91
With HRFE (3-D)	90.16	81.39	78.67	84.87	65.13	61.09
Δ ($\uparrow$)	+4.55	+6.87	+7.48	+7.85	+8.40	+8.18
Without HRFE (BEV)	91.04	84.35	83.17	82.04	61.61	57.46
With HRFE (BEV)	93.42	89.59	87.61	88.14	69.85	65.58
Δ ($\uparrow$)	+2.38	+5.25	+4.44	+6.10	+8.24	+8.12

F. Results

1) *Quantitative Results:* We evaluated our model on the official KITTI dataset for both 3-D and BEV-OD benchmarks. The results, as shown in Tables III and IV, highlight our model’s effectiveness compared to existing approaches. The improvement in the moderate and hard categories highlights the effectiveness of our backbone and choice of multiscale aggregation and spatial attention for enhanced feature representation. Fig. 9 presents the precision–recall curves for the car and cyclist categories, demonstrating the detection capabilities across different scenarios.

2) *Qualitative Results:* The visual results are shown in Fig. 10. We rendered the 3-D bounding boxes on the KITTI test samples for visual representation. To draw bounding boxes on LiDAR, we used the Open3D package, an efficient tool for 3-D data processing and visualization. Our model is completely LiDAR-based, but we provide both LiDAR-based and image-based visualizations of the same scene for better visual understanding.

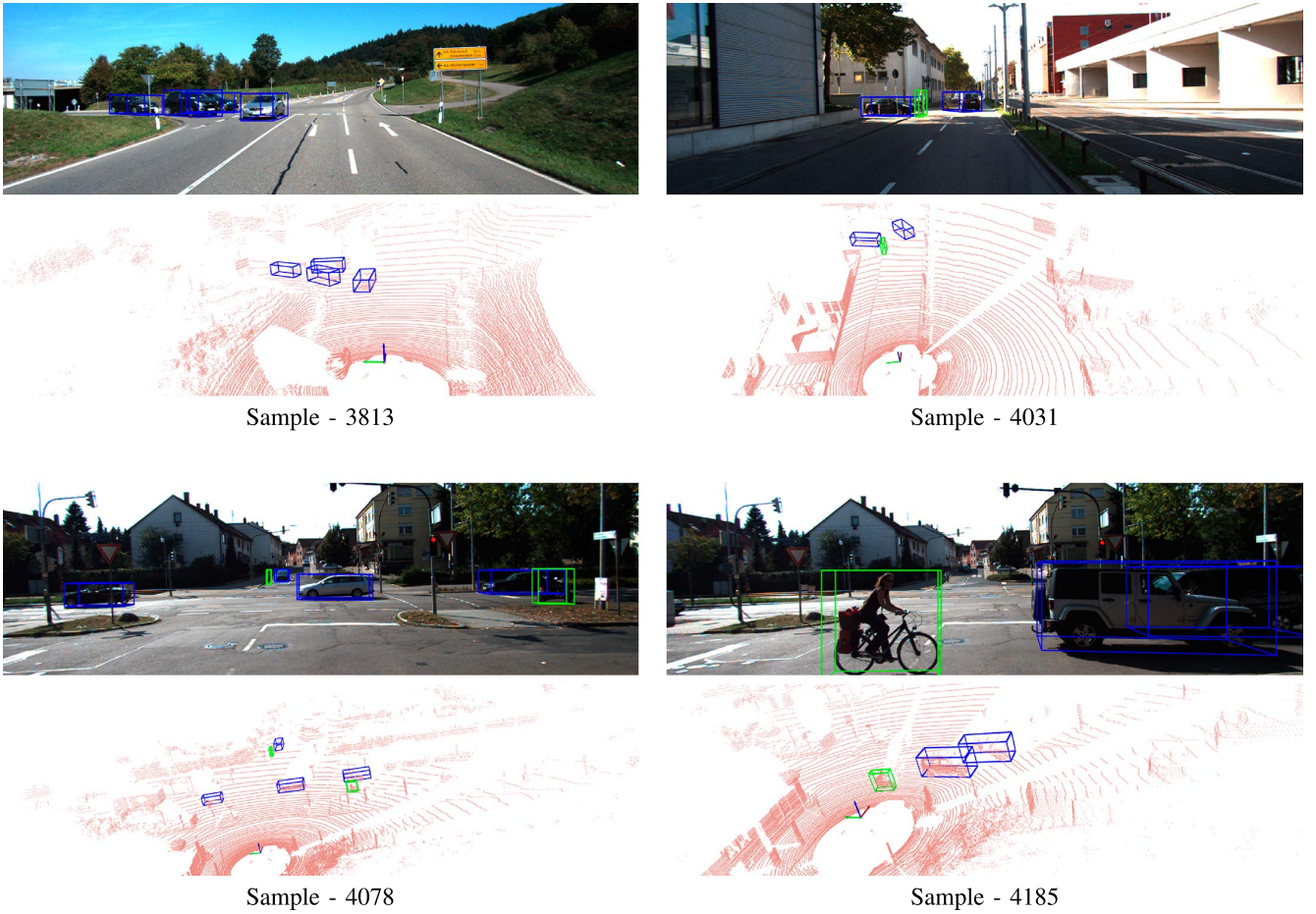


Fig. 10. Visualization of 3-D bounding boxes on KITTI test samples (3813, 4031, 4078, 4185). The camera (top) and LiDAR (bottom) views are shown for each of these samples. The bounding boxes are projected onto LiDAR point clouds using Open3D [53] and overlaid on the corresponding camera images for improved interpretability. This representation helps in analyzing object localization and alignment between different sensor modalities, providing deeper insights into model performance.

G. Ablation Studies and Analysis

1) *Effect of HRFE*: We evaluate the impact of the HRFE module in the proposed backbone by conducting ablation experiments on the KITTI validation split. The HRFE module is removed from the backbone to compare detection performance with and without it. Experiment results are reported in detail in Table V. The backbone with HRFE improves performance across all difficulty levels for both car and cyclist classes under 3-D and BEV detection. These improvements demonstrate that HRFE effectively enhances feature representation.

2) *Ablation Study on Multiscale Feature Aggregation*: To evaluate the effectiveness of multiscale feature aggregation, we conducted an ablation study comparing different receptive field sizes: Multiscale, 5×5 , 3×3 , and 1×1 . We used the KITTI validation split for this experiment. We conducted experiments to track the effect on 3-D car, 3-D cyclist, BEV car, and BEV cyclist detection tasks. Fig. 11 presents the results of the experiments and improvements over the average of all three single-scale variants across all difficulty levels. For 3-D car detection, our method achieves gains of +0.42, +0.39, +0.35, and +0.39 on Easy, Moderate, Hard, and mAP, respectively.

For 3-D cyclist, the improvements are more pronounced, with +4.34, +1.30, +1.10, and +2.25. In BEV car, the performance increases by +1.02, +0.42, +0.36, and +0.60, while BEV cyclist achieves +2.45, +0.24, +0.10, and +0.93 gains, respectively. The combination of different scales with our multiscale approach is giving better results, particularly in moderate and hard difficulty levels. The car class is stable or has a negligible impact in different categories, but the cyclist class gained a lot of improvement with the multiscale aggregation approach.

3) *Comparison of MSCA With FPN and ASPP*: We conducted experiments to evaluate the effectiveness of the MSCA module over established methods such as FPN and ASPP. We replaced the MSCA module with FPN and ASPP while keeping the remaining configuration identical. Table VI reports the results. All variants perform similarly on car detection with negligible differences, but MSCA shows improved results for the cyclist class. MSCA provides considerable accuracy with the fastest inference and lowest parameter count.

4) *Effect of DBFR Module*: We conducted experiments to report the impact of DBFR in the proposed pipeline. We conducted experiments on KITTI validation data. Table VII reports

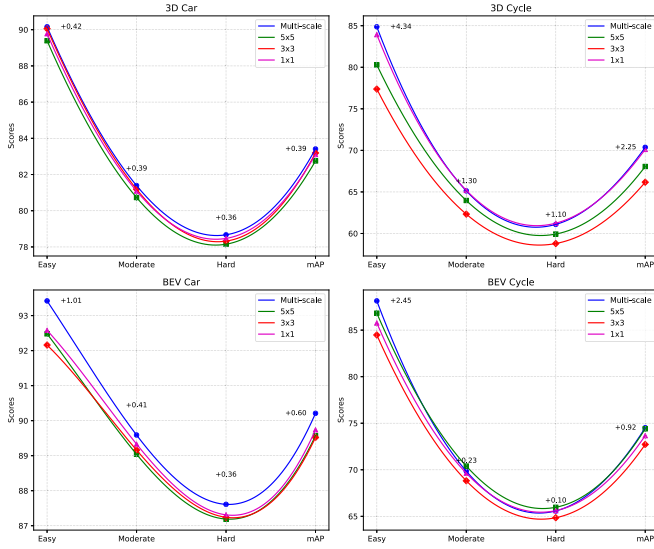


Fig. 11. Ablation study on multiscale feature aggregation. The multiscale approach consistently outperforms fixed receptive field sizes (5×5 , 3×3 , 1×1) across all evaluation settings, demonstrating its effectiveness in capturing multilevel spatial information.

TABLE VI
COMPARISON OF MSCA WITH FPN AND ASPP

Metric	With FPN	With ASPP	With MSCA (Ours)
3-D Car (mAP)	83.60	83.42	83.41
3-D Cycle (mAP)	67.91	69.67	70.37
BEV Car (mAP)	90.03	90.39	90.21
BEV Cycle (mAP)	73.97	73.84	74.53
Inference time (s)	0.0382	0.0568	0.0312
Params	3.65M	3.52M	3.44M
FPS	26	17	32

Note: Bold represents the top performing entry.

a comparison between models without DBFR, with a single-branch DBFR, and with the proposed dual-branch DBFR. The results are reported for both 3-D and BEV representations across car and cyclist classes. The dual-branch DBFR consistently provides better performance.

5) *Kernel Ablation Study*: We selected the kernel size in the spatial attention module by conducting experiments. We analyzed the impact of different kernel sizes 3×3 , 5×5 , and 7×7 . We conducted this experiment on the KITTI test set. Table VIII presents the results with different kernel sizes for 3-D and BEV detection of both car and cyclist classes. This suggests the kernel 3×3 most effectively captures local spatial features while larger kernels may introduce unnecessary complexity. These results highlight the importance of kernel size 3×3 in the spatial attention mechanism module of our backbone.

6) *Effect of CASA Module*: We conducted an ablation experiment to evaluate the impact of CASA block on 3-D and BEV detection for both cars and cyclists. Unlike existing methods that perform ablation on the validation set, we tested on the test set. Table IX presents the results with and without CASA block for 3-D and BEV detection on both car and cyclist classes. We can clearly see a significant improvement in the cyclist classes, where small object representation and occlusion handling are critical. For the car class, the performance remains comparable

TABLE VII
ABLATION STUDY ON DBFR FOR 3-D AND BEV DETECTIONS ON THE KITTI DATASET

Model	Car (AP)			Cyclist (AP)		
	Easy	Mod.	Hard	Easy	Mod.	Hard
Without DBFR (3-D)	89.49	80.98	78.29	80.79	63.98	60.40
With single branch (3-D)	90.50	81.25	78.55	78.62	62.68	59.61
DBFR (dual branch) (3-D)	90.16	81.39	78.67	84.87	65.13	61.09
Without DBFR (BEV)	91.89	88.03	87.37	85.46	68.93	65.07
With single branch (BEV)	93.30	88.97	87.37	83.13	67.87	64.16
DBFR (dual branch) (BEV)	93.42	89.59	87.61	88.14	69.85	65.58

Note: Bold represents the top performing entry.

TABLE VIII
EFFECT OF DIFFERENT KERNEL SIZES IN SPATIAL ATTENTION

Kernel	3-D Detection (AP)						BEV Detection (AP)					
	Easy	Mod	Hard	Easy	Mod	Hard	Easy	Mod	Hard	Easy	Mod	Hard
3×3	86.06	77.21	71.89	75.60	60.30	53.83	91.35	87.70	84.38	79.61	65.12	58.84
5×5	84.74	76.76	71.72	69.01	55.85	49.91	90.64	87.56	84.30	75.87	63.55	57.19
7×7	85.54	77.36	72.15	74.58	59.00	52.89	90.91	87.56	84.22	78.12	64.15	57.52

Note: The evaluation is performed on the KITTI test set. We conducted experiments for kernel sizes 3×3 , 5×5 , and 7×7 . Bold represents the top performing entry.

among different categories. The observations confirm the advantage of using spatial attention mechanism in the backbone.

7) *Effect of Pooling Strategies in CASA Block*: We conducted experiments to investigate the effect of pooling operations in the CASA block. We evaluated four configurations on the KITTI validation set: max pooling only, average pooling only, no pooling, and both poolings. As shown in Table X, using both poolings results in improved performance. The cyclist class performs best with both poolings in both 3-D and BEV metrics. This indicates that max and average pooling provide complementary spatial cues to enhance feature representation. Using either pooling method in isolation gives comparable accuracy for the car class. However, max pooling is slightly better for car and cyclist. Removing pooling entirely results in noticeable degradation, confirming the importance of pooling choices.

8) *Ablation Study on Attention Mechanisms*: To validate the effectiveness of the proposed CASA module, we conducted comparative experiments against established attention mechanisms such as Convolutional Block Attention Module (CBAM), squeeze-and-excitation (SE), and a transformer-based variant. We replaced our CASA block with these variants and used an identical configuration for all experiments. As shown in Table XI, although the SE-style attention shows a marginal improvement for the car-3D metric, the proposed CASA module delivers consistently better performance for the cycle-3D case as well as the BEV metrics. Further, the CASA module achieves the lowest inference time and the highest frame rate (FPS), while keeping the parameter count lower. Although the parameter count is close to that of SE-style attention, CASA shows a clear advantage over the other mechanisms. Overall, CASA provides a favorable balance between accuracy and efficiency, making it a strong alternative to heavier or more complex attention modules.

TABLE IX
IMPACT OF CASA ON 3-D AND BEV DETECTION

Model	Car (AP)			Cyclist (AP)		
	Easy	Moderate	Hard	Easy	Moderate	Hard
3-D Detection (Without CASA)	86.04	77.63	72.40	71.56	57.17	51.15
3-D Detection (With CASA)	86.06	77.21	71.89	75.60	60.30	53.83
Difference (Δ)	+0.02	-0.42	-0.51	+4.04	+3.13	+2.68
BEV Detection (Without CASA)	91.46	87.85	84.55	78.22	64.03	56.99
BEV Detection (With CASA)	91.35	87.70	84.38	79.61	65.12	58.84
Difference (Δ)	-0.11	-0.15	-0.17	+1.39	+1.09	+1.85

Note: The experiments were conducted on the KITTI test set.

TABLE X
EFFECT OF POOLING STRATEGIES IN THE CASA BLOCK

Metric	Max Pool	Avg Pool	No Pool	Both Poolings
Car (mAP)-3-D	83.56	83.55	83.15	83.41
Cyclist (mAP)-3-D	70.09	69.44	67.22	70.37
Car (mAP)-BEV	90.01	89.91	89.71	90.21
Cyclist (mAP)-BEV	73.59	73.37	71.78	74.52

Note: Bold represents the top performing entry.

TABLE XI
ABLATION STUDY TO COMPARE CASA MODULE WITH ESTABLISHED ATTENTION MECHANISMS

Model	3-D		BEV		Time/Memory		
	Car	Cycle	Car	Cycle	Inference (s)	Params	FPS
CBAM_style	83.28	66.94	90.16	70.59	0.0453	3 472 490	22
SE_style	83.43	67.20	89.59	72.52	0.0380	3 443 720	26
Transformer style	82.90	68.62	89.23	73.28	0.0356	3 972 104	28
Our CASA	83.41	70.37	90.21	74.52	0.0312	3 441 480	32

Note: Bold represents the top performing entry.

TABLE XII
COMPARISON OF PROPOSED BACKBONE WITH LIGHTWEIGHT ALTERNATIVES

Model	Inference Time	Params	3-D-Car (mAP)	3-D-Cyclist (mAP)	BEV-Car (mAP)	BEV-Cyclist (mAP)
EfficientNet-B0	0.0341	5 091 528	72.04	41.45	81.57	46.07
MobileNetV2	0.0543	3 441 032	74.12	40.22	82.74	44.74
ResNet (tweak)	0.0462	9 820 168	84.08	68.83	90.13	73.18
Ours	0.0312	3 441 480	83.41	70.37	90.21	74.52

Note: Bold represents the top performing entry.

9) *Comparison With Lightweight Backbone Alternatives:* We conducted an ablation study to compare our proposed backbone with well-known lightweight architectures such as MobileNetV2 [54], EfficientNet-B0 [55], and ResNet-18 [56]. We replaced our backbone with each while maintaining consistent pipeline integration and model configuration.

As shown in Table XII, our model achieves a good balance between accuracy and efficiency. While MobileNetV2 reports the lowest parameter count (3.44M), it comes with lower detection performance in the cycle class. EfficientNet-B0 offers competitive inference speed, but accuracy is limited in both 3-D and BEV. The ResNet-based variant shows comparative results, but it has larger parameters and slower inference time compared to our method. In contrast, our backbone maintains decent performance across most categories, especially in cycle detection. Our model has the lowest inference time (0.0312 s), indicating a better balance in performance and complexity.

V. CONCLUSION

The proposed model is a completely end-to-end single-stage LiDAR-based 3D-OD model. We proposed a novel backbone with multiscale aggregation and spatial attention mechanism and used a standard grid-based single-stage pipeline to implement the model. We reduced almost $\sim 40\%$ of the parameters in the proposed backbone. The proposed model achieved competitive accuracy by being lightweight. Our single-stage model outperforms many existing single-stage models and achieves comparable results over two-stage models. We evaluated our model using the official KITTI test server, and our results are featured on the official leaderboard of KITTI. We conducted extensive ablation experiments to analyze the impact of our backbone architecture and its individual components. The experimental results strongly support the efficiency of our backbone.

The current evaluation focuses on the car and cyclist classes, as pedestrian objects exhibit sparse and highly variable LiDAR point representations, which pose additional challenges for reliable detection in purely LiDAR-based single-stage frameworks. Extending the evaluation to include the Pedestrian class remains an important direction for achieving a more comprehensive assessment. In addition, all experiments in this work are conducted on the KITTI dataset. Although KITTI is a widely used benchmark, the proposed backbone itself is not dataset-specific and is designed to be lightweight and modular. With minor configuration-level adaptations, it can be integrated into detection pipelines for larger scale datasets such as nuScenes and Waymo.

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